Today’s Plan

Career Spotlight

Web Development Workshop

Career Spotlight

Visit menti.com and use code 2499 8114

ACCESS Operations: NSF #2138307

Visit menti.com and use code 2499 8114

What's one interest or hobby you have that has nothing to do with CS or engineering?

Hello World, My Name Is

```html
<h1>Chun Chan</h1>
<h2>Research Software Developer</h2>
<h3>Indiana University</h3>
```

T | Games

Create four groups of four!

THINGS THAT ARE YELLOW
BUTTER, PIKACHU, RUBBER DUCK, SCHOOL BUS

BILLIARDS TERMS
BREAK, CUE, POCKET, RACK

KINDS OF WOOD PLUS "S"
SASH, SOAK, SPINE, STEAK

SLANG FOR A SAILOR
JACK, SALT, SEA DOG, TAR

Words before __ knife

Career Built on Connections

History

- Undergraduate in **Music Technology** in Malaysia
  - Alongside the tech curriculum: choir, music history, music theory
- Had some idea that music & tech were linked in some way
  - Audio production and engineering was moving from analog to digital
  - mp3s becoming more prevalant
  - Personal listening devices like the iPod were taking shape
- Accepted into **Northwestern University** for a Master's in Music Technology

What I Studied vs. What It Secretly Was

Music Technology

- Sound Synthesis
- Signal Processing
- Recording & Mixing
- Music for Film & TV

Actually...

- Algorithm design & DSP
- The math behind digital audio
- Data pipelines & quality control
- Communicating complex ideas clearly

> *The arts and sciences are less separate than you might think.*

The Pivot

My first lab job at Northwestern was studying live pitch-shift modulation of speech.

I already knew this — it was audio engineering.

- That connection opened doors to adjacent labs
  - Hearing loss & hearing aids
  - Neurodiverse speech production & perception
  - Linguistics & sociolinguistics

Language & Music are highly connected!

- When COVID hit, everything moved online
  - In-person experiments → remote web experiments
  - My role expanded from lab technician to full-stack developer

What I Actually Do

Research Software Developer — Northwestern University → Indiana University

Creative problem solving: helping researchers get from point A to point B.

- Translate research questions into working software
- Design experiments that run in browsers, servers, and labs
- Bridge the gap between *"I need to study this"* and *"here's how we implement & measure it"*
- Guide students through the full research lifecycle

> *Technical skill is only part of it. The rest is communication, curiosity, and creativity.*

A Career You Probably Haven't Heard Of

Almost every research lab needs someone like this — and very few people know it exists.

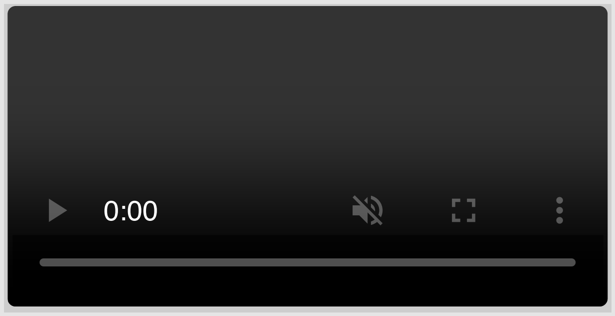
- **Every field of academic research runs on software and data**
  - Psychology, linguistics, biology, political science, neuroscience...
- Researchers are domain experts, not necessarily developers
- Someone has to build the tools, run the experiments, and wrangle the data

What makes you valuable isn't mastering a specific stack. It's being able to pick up whatever the research demands quickly.

> *It's not React vs. Vue, or Python vs. R. It's: can you learn what you need, when you need it?*

See It in Action

Self-paced reading experiment — participants press space to reveal each word



The Range

Building & Running

- Web experiments (jsPsych, PCIbex, Gorilla, Psycopy)
- HPC pipelines for large-scale data
- Lab audio recording & equipment
- IRB documentation & participant logistics

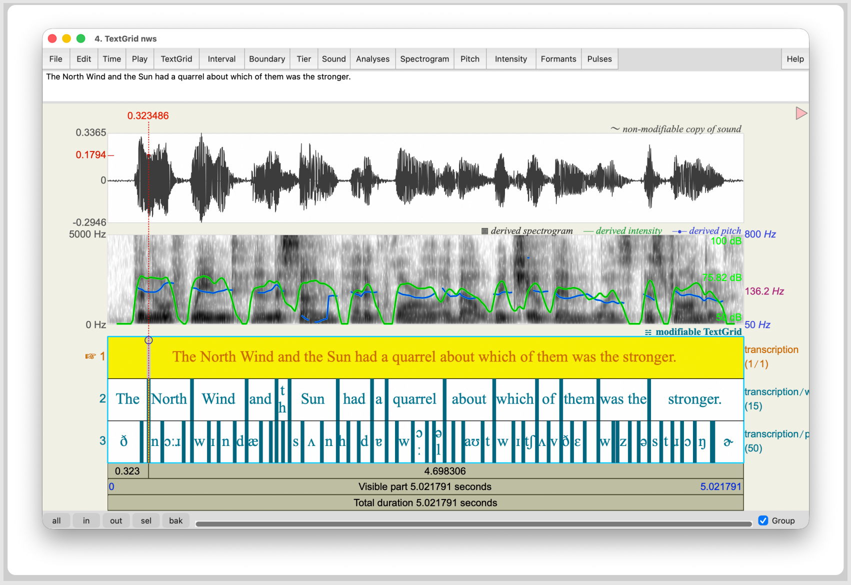
Analyzing & Discovering

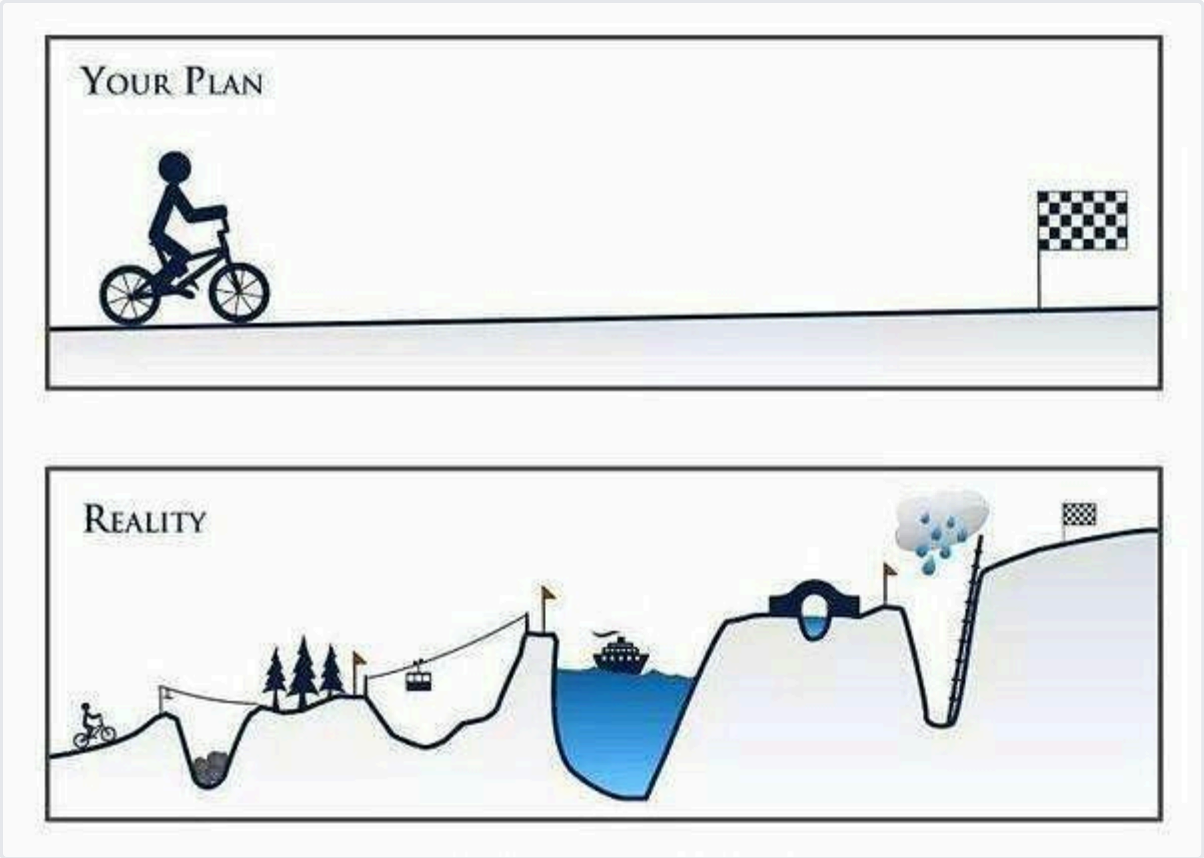
- Speech audio: spectrograms, forced alignment
- Speech recognition on noisy & non-native speech
- Sociolinguistics: vowel shifts, code switching
- Neurodiverse speech production & perception

Some Unsolved Problems

- Speech-to-text systems erase disfluencies — the very thing some researchers study
- ASR models still struggle with child speech and non-native accents
- How do you measure attention in a remote participant you've never met?
- How do you study natural speech when the study itself makes speech feel unnatural?

> *Somebody has to figure these out. It helps to be someone who understands both the science and the craft.*




YOUR PLAN
REALITY

The Path I Didn't Plan

Music → Audio Engineering → Research Software → still figuring it out

Your Path

CS/Engineering/Math/Physics → ? → ? → ?

You don't have to have it figured out yet. Be open to making connections and stay curious.

The overlap of what you love and what the world needs, that's where the interesting problems live.